

we will search through to find a particular arcsin value. The `sin` function is periodic and takes on all of its possible values between $-\pi/2$ and $\pi/2$, so numbers in between those extremes will constitute our domain. Next, we calculate sine values for each value in the domain. We call `binarysearch()` to find the position of the number whose sine is the number we're looking for, and return the domain value with the corresponding index, like so:

```
def inverse_sin(number):
    domain = [x * math.pi/10000 - math.pi/2 for x in list(range(0,10000))]
    the_range = [math.sin(x) for x in domain]
    result = domain[binarysearch(the_range,number)]
    return(result)
```

You can run `inverse_sin(0.9)` and see that this function returns the correct answer: about 1.12.

This is not the only way to invert a function. Some functions can be inverted through algebraic manipulation. However, algebraic function inversion can be difficult or even impossible for many functions. The binary search method presented here, by contrast, can work for any function, and with its $O(\log(n))$ runtime, it's also lightning fast.

Summary

Sorting and searching may feel mundane to you, as if you've taken a break from an adventure around the world to attend a seminar on folding laundry. Maybe so, but remember that if you can fold clothes efficiently, you can pack more gear for your trek up Kilimanjaro. Sorting and searching algorithms can be enablers, helping you build newer and greater things on their shoulders. Besides that, it's worth studying sorting and searching algorithms closely because they are fundamental and common, and the ideas you see in them can be useful for the rest of your intellectual life. In this chapter, we discussed some fundamental and interesting sorting algorithms, plus binary search. We also discussed how to compare algorithms and use big O notation.

In the next chapter, we'll turn to a few applications of pure math. We'll see how we can use algorithms to explore the mathematical world, and how the mathematical world can help us understand our own.